Available online at www.sciencedirect.com
ScienceDirect
Proceedings of the Combustion Institute 33 (2011) 175–183
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

Oxidation kinetics of *n*-nonane: Measurements and modeling of ignition delay times and product concentrations

B. Rotavera [a,*], P. Diévart [b], C. Togbé [b], P. Dagaut [b], E.L. Petersen [a]

[a] *Department of Mechanical Engineering, Texas A&M University, College Station, TX 77843, USA*
[b] *Centre National de la Recherche Scientifique (CNRS), Institut de Combustion Aérothermique Réactivité et Environnement (ICARE) 1c, Ave Recherche Scientifique, 45071 Orléans Cedex 2, France*

Available online 21 August 2010

Abstract

Oxidation of *n*-nonane (n-C_9H_{20}) under conditions of high dilution (>97% inert) has been studied over a broad range of temperature ($530 < T\,(K) < 1591$) and equivalence ratio ($\phi = 0.5, 1.0, 2.0$) at pressures near 1 and 10 atm using shock-tube and jet-stirred reactor facilities. Excited-state hydroxyl radical (OH*) time histories were measured using emission spectroscopy of the $A^2\Sigma^+ \rightarrow X^2\Pi$ transition of OH behind reflected shock waves from which ignition delay and peak formation times were extracted. Temperature-dependent species concentrations were measured using gas chromatography, FTIR, TCD, and FID of jet-stirred reactor combustion products. Ignition delay times show a strong dependence on equivalence ratio, increasing by a factor of nearly 5 at both 1 and 10.4 atm. An overall ignition delay time correlation was constructed, revealing a pressure dependence of $P^{-0.48}$. Experimental data from both facilities were utilized to develop and validate a chemical kinetics mechanism for *n*-nonane oxidation. Kinetic model predictions of ignition delay time compare well, particularly for the lean and stoichiometric mixtures. Jet-stirred reactor data show excellent overall agreement with major species, as well as alkanes and alkenes present in the combustion products. Alkenes up to C_9 were produced from *n*-nonane oxidation and ethylene, a dominant product of *n*-nonane thermal decomposition, is identified as the most abundant among them. The present study provides an extensive series of fundamental measurements on *n*-nonane oxidation, resulting in the formulation of a mechanism used to describe and predict associated reaction kinetics.
© 2010 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

Keywords: *n*-Nonane oxidation; Kinetics mechanism; Shock-tube; Jet-stirred reactor; Ignition

1. Introduction

Fundamental chemistry measurements of higher-order hydrocarbons are a necessity in developing adaptive chemical kinetics models which benefit from comprehensive studies spanning a large parameter space, including temperature, pressure, equivalence ratio, fuel concentration, and dilution level. Fundamental data on *n*-alkanes from methane to octane are abundant, and combustion-related measurements on *n*-decane, *n*-dodecane, and other heavy hydrocarbons are also available, although to a lesser extent. Studies on *n*-nonane oxidation chemistry are, however, scarce. Normal

* Corresponding author. Fax: +979 845 3081.
E-mail address: Rotavera@tamu.edu (B. Rotavera).

1540-7489/$ - see front matter © 2010 The Combustion Institute. Published by Elsevier Inc. All rights reserved.
doi:10.1016/j.proci.2010.05.055

alkanes compose a significant proportion of diesel and aviation fuels with jet fuels containing alkanes at levels of 50–70% by volume [1]. *n*-Nonane is present in petroleum-based fuels [2], and it is thus of interest to treat the straight-chained alkane experimentally with the aim of building a comprehensive kinetics mechanism to describe its oxidation over a wide range of thermodynamic conditions.

Fundamental studies on the pyrolysis and oxidation of n-C_9H_{20} are limited [3–8], yet it is of interest to consider these processes to elucidate important chemical physics and reaction kinetics that help describe ignition and combustion mechanisms of such higher-order hydrocarbons.

Kunzru et al. employed a flow reactor to investigate the thermal cracking of *n*-nonane and compared experimental results to kinetic theories of Rice and Woinsky [3]. Product distributions were measured as a function of *n*-nonane conversion percentage and temperature using gas chromatography. Over the range of temperature covered in the study, 920–1020 K, ethylene (C_2H_4) production from *n*-nonane decomposition was shown to be considerably higher than any other hydrocarbon species considered.

An overlapping study on the pyrolysis of straight-chained hydrocarbons from C_9 to C_{22} also showed, at 9.3% decomposition by weight, olefins to be predominant in the product yield of *n*-nonane thermal decomposition near 900 K; notable within the yield, a significant formation of ethylene [4]. Measurements of the rate coefficient of n-C_9H_{20} + OH were obtained at high temperature (1100 K) using a shock-tube facility [5]. A second-order rate coefficient for consumption of OH by *n*-nonane of 4.55×10^{-11} cm^3 mol^{-1} s^{-1} resulted from a series of experiments spanning a wide range of *n*-nonane concentration.

A recent and expansive work on *n*-alkane reactivity by Westbrook et al. covered hydrocarbons from C_8 to C_{16} in which a kinetics mechanism was developed using a plethora of experimental data [6]. The model is shown to agree well with *n*-nonane pyrolysis measurements of Zhou et al. [4]. Model computations were carried out for stoichiometric ignition delay times of the group of higher-order alkanes at 13.5 bar, where overlap was observed at temperatures higher than 1000 K. Due to differences in chain length, the hydrocarbons displayed some variation in ignition delay times over certain regions of temperature. Within the negative temperature coefficient region, 890–960 K, variance in ignition times amongst the hydrocarbons of as much as 1 ms is reported. Near 800 K, the fuels differ at most in ignition delay time by a measure of 600 μs. Ignition delay times remained distinguishable as temperature decreased to ~700 K, below which the ignition behavior amongst the various fuels again overlaps. A kinetic model (JetSurF) was constructed for prediction of laminar flame speeds and extinction strain rates of C_5–C_{12} *n*-alkanes in air by Ji et al. [7]. Model calculations were compared to experimental data at 1 atm, where deviation from flame speed measurements was highest (2.5 cm/s) for *n*-nonane–air mixtures from $0.9 < \phi < 1.1$. Davidson et al. [8] reported measurements of *n*-nonane ignition delay times using a heated shock-tube. Fuel-lean and stoichiometric conditions were utilized where oxygen concentration remained fixed at 4% and pressure was maintained near 2.0 or 3.5 atm.

There remains a need for measurements of higher-order alkane oxidation processes under both dilute and non-dilute conditions with complementary kinetics modeling. Presented herein is a wide-ranging study of *n*-nonane oxidation at two pressures, 1 and 10 atm, covering fuel-lean, stoichiometric, and fuel-rich equivalence ratios over a broad range of temperature and a detailed mechanism describing the reaction kinetics. Shock-tube and jet-stirred reactor (JSR) facilities were employed as described below, and all experiments were performed under highly dilute conditions (⩽0.1% fuel) to minimize unsteady thermodynamic conditions during chemical reaction. The authors report in the present work an extensive treatment of low- and high-temperature *n*-nonane oxidation and propose a detailed mechanism describing the chemical kinetics.

2. Experimental approach and kinetic model development

Experiments on highly dilute, gas-phase mixtures of n-$C_9H_{20}/O_2/Ar$ were performed using shock-tube [9,10] and jet-stirred reactor [11] facilities. Three different mixture compositions were studied (Table 1) covering fuel-lean ($\phi = 0.5$) to fuel-rich ($\phi = 2.0$) conditions under low (~1 atm) and intermediate (~10 atm) pressures over the temperature range $530 < T\,(\mathrm{K}) < 1220$ for the JSR and $1268 < T\,(\mathrm{K}) < 1627$ for shock-tube experiments. High-purity gases (Ultra High Purity O_2, Ar, and N_2) and n-C_9H_{20} with a purity level of >99% (Sigma–Aldrich N29406) were used for all JSR and shock-tube experiments.

2.1. Chemiluminescence and ignition delay time measurements

Ignition delay times and species time histories from the oxidation of high-purity n-$C_9H_{20}/O_2/Ar$ mixtures were measured using two different shock-tube facilities. The low-pressure shock-tube described by Rotavera and Petersen [9] was employed for the 1.5-atm measurements, while the high-pressure shock-tube facility described by de Vries et al. [10] provided overlapping low-pressure as well as higher-pressure data. All experiments were performed behind reflected shock

Table 1
Molar composition of mixtures (Ar used in shock-tube mixtures, N_2 used in JSR mixtures).

ϕ	n-C_9H_{20}	O_2	Ar (N_2)
0.5	0.001	0.028	0.971
1.0	0.001	0.014	0.985
2.0	0.001	0.007	0.992

waves, where test temperature and pressure were determined by measuring the incident shock speed. An average attenuation rate of 1% per meter in each facility imposes an uncertainty in temperature of ± 10 K, as documented in Petersen et al. [12].

Ignition delay time is defined herein using the method of steepest ascent where the intersection of lines drawn along the steepest rate-of-change of OH* de-excitation and a horizontal which defines the zero-concentration level (Fig. 1), is measured relative to time-zero (formation of reflected-shock conditions).

Incipient OH* formation near time-zero, observed slightly in Fig. 1, was present in the majority of shock-tube experiments. The behavior showed a marked increase in magnitude with increasing equivalence ratio and temperature, while increased pressure mitigated the effect. Model predictions of OH*(t) also reflect this first OH* peak near time-zero.

Chemiluminescence profiles for OH* were measured for each mixture at average pressures of 1.5 and 10.4 atm, where UV emission exiting a CaF_2 window through a 400-μm slit was captured using a photomultiplier tube (Hamamatsu 1P21). A narrow-bandwidth bandpass filter (307 ± 10 nm) was used for measuring radiation from time-dependent OH* de-excitation to reduce the potential for interference from other emitters.

In each of the abovementioned mixtures utilized for shock-tube experiments, careful attention was paid to the issue of possible fuel condensation

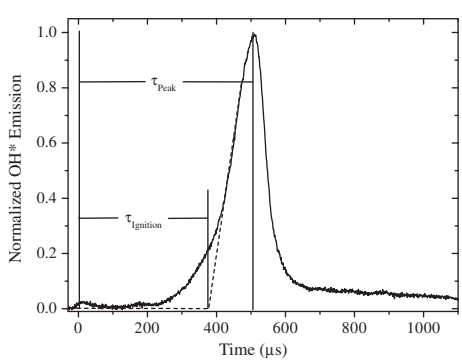

Fig. 1. Normalized OH* emission profile for 0.001n-$C_9H_20 + 0.014O_2 + 0.985Ar$ ($\phi = 1.0$). $\tau_{Ignition} = 378$ μs, $\tau_{Peak} = 508$ μs, $T_5 = 1447$ K, $P_5 = 1.57$ atm.

and subsequent compromising of mixture composition. The partial pressure of the n-nonane was kept below 60% of its saturation vapor pressure at room temperature (~3.6 torr), significantly reducing the potential for condensation inside of the mixing tank due to any ambient temperature fluctuations and/or during the introduction of the fuel vapor and subsequent gases. To enhance mixing, turbulence was promoted during mixture preparation where the constituent gases (O_2, Ar) are entered into the mixing chamber (containing vapor-phase fuel) through a long tube that extends the entire inside length of the tank. Gases expel outward through numerous 800-μm-diameter holes which are staged 90° out of phase from one another, thereby inducing turbulent mixing. Further, mixtures remained stagnant for a period of at least 24 hrs prior to use, allowing for diffusion processes to further ensure homogeneity. These three precautions provided confidence in the composition of the reactive mixture entering the shock tube. The adequacy of these methods was further proven by repeating the mixtures several times each, with consistent results each time, often repeating tests and mixtures in both shock-tube facilities. A remaining uncertainty potential was condensation of the n-nonane undergoing expansion as it enters the shock tube. Initial pressure levels in the shock tube were maintained for several minutes, and no change in pressure was observed.

Non-ideal boundary layer effects, present to some extent in all shock-tube experiments, had a negligible impact on the results presented herein. To elaborate, the measured change in pressure (dP/dt) behind the reflected shock wave for the higher-pressure, 15.24-cm diameter tube was less than 2% per ms for all experiments, based on argon bath-gas tests; the corresponding worst-case increase in pressure for the lower-pressure, 10.8-cm square tube was slightly higher, yet less than 3% per ms. The corresponding increase in temperature for these dP/dt levels would be less than 15 K for the longest ignition delay times reported herein. Further evidence for the minimal effect of facility-related test condition uncertainties comes from the overlap in data obtained from the two different shock tubes, wherein several low-pressure experiments were performed in both facilities, with no difference between them within the routine uncertainty in the experimental method. Experimental uncertainty in reported ignition delay times, approximately 10%, is ascribed mainly to uncertainty arising in measurement of shock speed which is subsequently used for calculation of temperature behind the reflected shock.

2.2. Product-species concentration measurements

Temperature and pressure dependencies of product-species concentration produced from the

oxidation of n-$C_9H_{20}/O_2/N_2$ mixtures were measured using a fused-silica jet-stirred reactor with an internal volume of 30 cm^3 described by Dagaut and Cathonnet [11]. The mixture was preheated to a given initial temperature to set reaction temperature and avoid condensation of the low-vapor-pressure alkane and then injected into the reactor body using four nozzles (i.d. = 1 mm). Turbulence is induced prior to entry into the reactor body, thus homogeneity of conditions through enhanced mixing (jet-stirring) of the already turbulent gases is attained. The high dilution level (>97% diluent) reduces temperature gradients and the release of energy from oxidation reactions. Residence times within the reactor remained fixed at 70 and 700 ms for 1- and 10-atm conditions, respectively.

A vertically adjustable probe positioned along the centerline of the reactor provided assurance of temperature uniformity (~1 K/cm) and uniformity of the gaseous mixture. Fourier-transform infrared spectroscopy (FTIR), flame ionization detection (FID), and thermal conductivity detection (TCD) were employed to measure species concentrations. Concentration analysis of select product species was performed during the experiment using FTIR in a 413 K, temperature-controlled gas cell after which combustion products (~40 mtorr) were captured using a quartz probe and collected in Pyrex bulbs for immediate analysis using gas chromatography in concert with either FID or TCD.

2.3. Kinetic mechanism for n-nonane oxidation

Kinetic modeling was performed using the CHEMKIN computer package: the JSR results were computed with the PSR code, and shock-tube experiments with the SENKIN code. The proposed kinetics mechanism, derived from a mechanism previously developed for n-decane oxidation [13], has a strong hierarchical structure and was designed to model both low- and high-temperature oxidation. The low-temperature n-nonane subset involves the formation of nonylperoxy radicals and derived reactions (isomerization, cyclization, peroxidation, etc., to form cyclic ethers, ketohydroperoxides, and aldehydes). Rate coefficients mainly come from Buda et al. [14]. The high-temperature reactions primarily consist of C–C and C–H bond scissions from n-nonane and nonyl radicals. To enable shock-tube ignition delays and the OH*-profile modeling, the OH* subset proposed by Hall and Petersen [15] was utilized. The presented mechanism, available in the supplemental content, consists of 910 species and 3755 reactions (most of them reversible).

Thermodynamic data were taken from the Burcat and Ruscic database [16] when available, or were estimated using the THERGAS software [17] based on group additivity methods proposed by Benson [18]. Rate coefficients for reverse reactions were computed from the corresponding forward rate coefficients and the appropriate equilibrium constant, $K_c = k_{Forward}/k_{Reverse}$, calculated from thermochemical data.

3. Jet-stirred reactor results

Temperature-dependent concentrations for 25 products of combustion were measured for each mixture and condition using sonic probe sampling coupled with the abovementioned diagnostics. Temperature was varied from 530 K to a peak temperature of 1220 K by adjusting the flow rate, while pressure was maintained at either 1 or 10 atm. Carbon-balance calculations were performed to compare the carbon atoms measured in the combustion products to those predicted by the mechanism. Under 1 atm pressure, kinetic model predictions are within c.a. 10% of the measured carbon; at 10 atm, the error is less than 2%. Mole fraction measurements of CH_4, C_2H_4, n-C_9H_{20}, CH_3OH, CH_2O, CO, CO_2, H_2O, and H_2 from stoichiometric n-nonane oxidation, representative of the overall data set, are presented for both 1 and 10 atm in Fig. 2; qualitative behavior is consistent with varying equivalence ratio.

Discussed in the ensuing sections are trends in mole fractions as a function of equivalence ratio, temperature, and pressure with specific attention paid to major species, oxygenates, and olefins. Contained in the supplemental content are figures referenced in the sections below with plots of oxygenated hydrocarbons and H_2O/H_2 species profiles.

3.1. Fuel-lean condition ($\phi = 0.5$)

At 1 atm pressure, trends of all major species (CO_2, CO, H_2O) are captured well by the model and, quantitatively, CO_2 and H_2O are accurately modeled. A slight under-prediction of H_2O is evident near 1070 K, and between the temperature range $1040 < T\,(K) < 1100$ CO is under-predicted by the model.

Oxygenated species present at 1 atm include formaldehyde (CH_2O), methanol (CH_3OH), acetaldehyde (CH_3CHO), 2-propenal (C_3H_4O), and acetone (C_2H_5CHO) and were measured mainly at temperatures in excess of 900 K with some presence also detected for the temperature range $620 < T\,(K) < 680$. For a reaction pressure of 10 atm, in addition to the aforementioned products, 2-butanone (C_3H_7CHO) and 2-pentanone (C_4H_9CHO) were also identified as part of the product composition. The increase in pressure to 10 atm led to the formation of these species over a broader temperature range and a larger mole fraction at a given temperature.

Olefins up to C_9 were detected at both pressures. Ethylene, at high temperatures (>950 K),

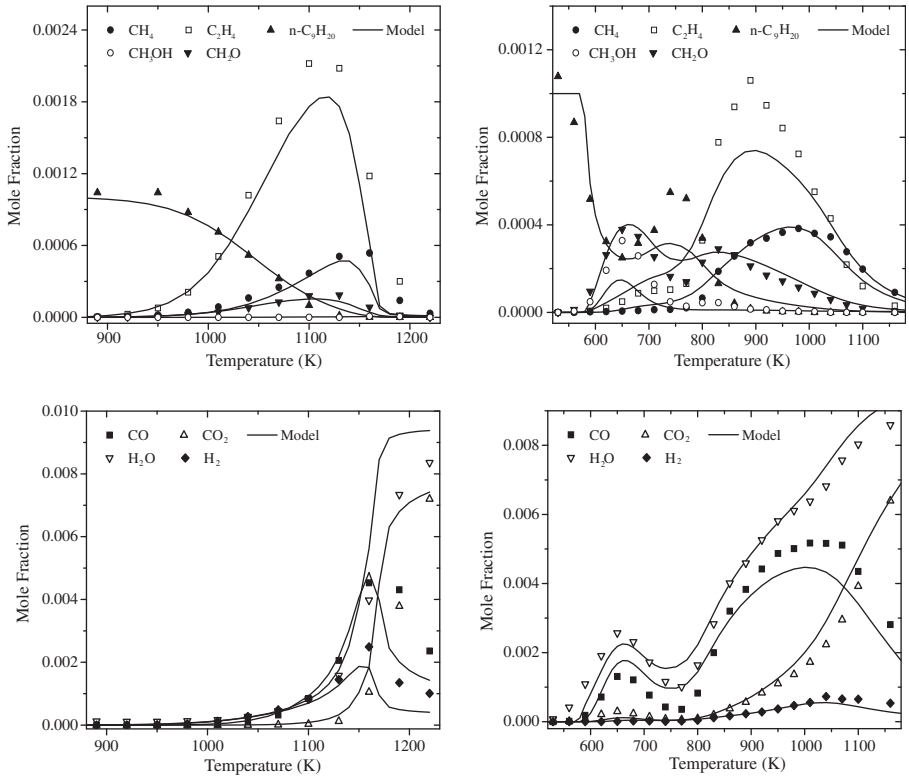

Fig. 2. Comparison of measured mole fractions to computed results using the present mechanism for 0.001*n*-$C_9H_2O + 0.014O_2 + 0.985N_2$ ($\phi = 1.0$) at 1 atm (Left) and 10 atm (Right); Methanol is scaled by a factor of two. Residence times: 70 ms (1 atm), 700 ms (10 atm).

exceeded the concentration levels of other olefins at 1 atm; at higher pressure, the presence of the species was broadened, covering the entire range of experimental temperature. Ethylene appeared as the dominant olefin for all temperatures at 10 atm as well as for 1 atm at temperatures where the species was present.

Qualitatively, trends of all species are predicted rather well; however, peak temperatures are under-predicted for minor species including propene (C_3H_6), butene (C_4H_8), pentene (C_5H_{10}), and acetaldehyde. An increased consumption of *n*-nonane at high pressure is noteworthy, as is the rapid decrease in ethylene at temperatures greater than 1050 K (1 atm) and 900 K (10 atm).

3.2. Stoichiometric condition ($\phi = 1.0$)

At both pressures, trends of CO_2, CO, and H_2O are reproduced accurately at the stoichiometric condition; however, deviations are present at temperatures near 1200 K for the 1-atm condition. At 10 atm, CO is slightly under-predicted above 900 K.

Oxygenated species were present mainly at higher temperatures, at 1 atm, where most of the oxygenates were suppressed below 1040 K. Formaldehyde appears over the range of temperature covered, from 2.9 ppmv (920 K) to 18.3 ppmv (1190 K). At higher pressure, high-order species up to C_4H_9CHO are formed, however, oxygenated species cease at temperatures in excess of 1040 K, with the exception of formaldehyde which is present at 4.6 ppmv at the peak temperature (1160 K).

Ethylene again appears as the dominant olefin at both 1 and 10 atm. Peak concentration for C_2H_4 occurs at 1100 K (2120 ppmv) for 1 and at 890 K (1060 ppmv) for 10 atm. For both conditions of pressure, as temperature increases a corresponding decrease in higher-order alkenes was noted. Model computations of ethylene mole fractions slightly under-predict experimental data at 10 atm, where increased pressure is observed to suppress its formation.

Trends of CH_4, C_2H_4, *n*-C_9H_{20}, CO, and CO_2 at both 1 and 10 atm are similar to those in the fuel-lean case; however, both CO and CO_2 are present and peak at lower temperatures. *n*-Nonane concentration is predicted well at both pressures, except for a slight under-prediction between 700 and 800 K at 10 atm where negative temperature coefficient (NTC) behavior is exhibited. Complete

consumption of the fuel occurs for $T > 1130$ and $T > 900$ K for 1 and 10 atm, respectively.

3.3. Fuel-rich condition ($\phi = 2.0$)

Carbon dioxide, CO, and H_2O are predicted extremely well at the rich equivalence ratio at both 1 and 10 atm; water is predicted particularly well by the model, including two points of inflection between 600 and 800 K where concentration rapidly decreases from 1740 ppm at 650 K to 671 ppm at 770 K, then rising to 2070 ppm at 830 K.

With the exception of CH_2O, oxygenated species were nearly non-existent at 1 atm. Presence of both acetone and acetaldehyde was detected between 1100 and 1130 K, however, only at small concentration (<7 ppmv). Formaldehyde reached peak concentration at 1160 K (121 ppmv). Similar behavior to that observed at other equivalence ratios was exhibited at 10 atm, namely the decreasing presence of higher-order oxygenates with increasing temperature. For both levels of pressure and over the entire range of temperature, C_2H_4 concentration was the highest of the olefins and increased with increasing temperature; ethylene behavior is captured well with peak yield occurring at 1150 and 980 K for 1 and 10 atm, respectively.

4. Shock-tube results

Due to diminutive energy release, ignition delay times ($\tau_{Ignition}$) are emission-based rather than pressure-based, using excited-state hydroxyl time histories and were measured for equivalence ratios of $\phi = 0.5$, 1.0, and 2.0 at pressures near 1.5 atm and 10.4 atm over a range of temperature $1176 < T\,(K) < 1591$. A complete table of ignition delay and peak OH* yield times is provided in the supplemental content. Ignition delay times for each condition are compared in the following sections against predictions made using the present kinetic mechanism, which shows strong agreement overall. Although not shown, the present data were also simulated using the mechanism of Westbrook et al. [6] and JetSurF [7], both showing reasonable agreement. Further, the model presented by the authors was employed to simulate *n*-nonane ignition data of Davidson et al. [8] and shows strong agreement. Refer to the supplemental content for abovementioned comparisons.

4.1. Fuel-lean ignition delay times

Measurements of ignition delay times at $\phi = 0.5$ were taken under two conditions of pressure, 1.5 and 10.4 atm, over temperature ranges $1285 < T\ (K) < 1590$ and $1176 < T\ (K) < 1401$, respectively (Fig. 3).

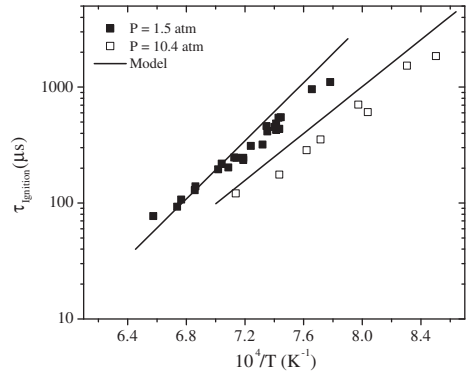

Fig. 3. Measurements and model predictions of fuel-lean ignition delay times. $0.001n\text{-}C_9H_{20} + 0.028O_2 + 0.971Ar$ ($\phi = 0.5$).

At the low-pressure condition, ignition delay times above 1340 K are accurately predicted by the model, however, calculations of ignition time for temperatures below 1340 K are slightly longer relative to the experimental trend, where a maximum disparity of 40% is present at 1285 K. Computation of ignition delay times at an average pressure of 10.4 atm agree well for low- and high-temperature measurements, deviating from the experimental trend by a peak measure of 32% below 1400 K.

Measured activation energy for the 1.5-atm trend ($E_A = 47.6$ kcal/mol) proved higher than that at higher pressure ($E_A = 42.4$ kcal/mol). The fuel-lean mixture presents the lowest activation energy at both pressures when compared to the other mixtures. Activation energy calculations using the model differ by 9.7 and 3.8 kcal/mol relative to 1.5 and 10.4 atm measurements, respectively.

4.2. Stoichiometric ignition delay times

Stoichiometric measurements, with temperature ranges of $1392 < T\ (K) < 1591$ for 1.5 atm pressure and $1287 < T\ (K) < 1460$ for 10.4 atm pressure, agree well at low pressure with predictions made using the present model, however, at higher pressure the calculated ignition times are longer than those measured experimentally by a factor of approximately 1.5 (Fig. 4).

At 1.5 atm, measurements taken at high temperature are predicted well by the model. Calculations at temperatures exceeding 1460 K are accurate within 25% of the measured experimental trend. Differences become evident as temperature is decreased below 1460 K, although only by a maximum deviation of 36%. An over-prediction by the model is apparent at high pressure over the range of temperatures covered; however, the behavioral trend of the high-dilution mixture is captured with a high degree of accuracy. The devi-

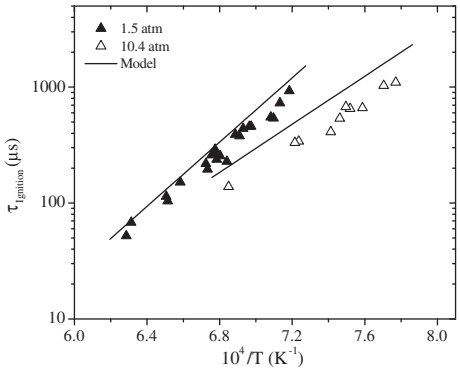

Fig. 4. Measurements and model predictions of stoichiometric ignition delay times. $0.001n\text{-}C_9H_{20} + 0.014O_2 + 0.985Ar$ ($\phi = 1.0$).

ation between model and experiment remains constant with changing temperature with a difference of 53% relative to the experimental data.

The stoichiometric mixture possessed the highest activation energy at the lower-pressure condition (59.0 kcal/mol) and the second highest at the higher pressure (44.9 kcal/mol). The decrease in activation energy of 14.1 kcal/mol with increasing pressure was the highest amongst the mixtures. Model calculations agree with measured ignition activation energy for 1.5 and 10.4 atm pressure, producing only slightly higher deviations of 63.4 and 47.4 kcal/mol, respectively.

4.3. Fuel-rich peak OH* formation times

Chemiluminescence profiles of excited-state species provide several characteristic and useful targets for validation of chemical kinetics mechanisms, in addition to ignition times. The time to peak formation of a species serves as a pertinent metric for assessing qualitative, time-dependent model behavior and in cases where ignition times agree, quantitative comparison of the chemical kinetics involving the species. Measurements of peak OH* formation times (τ_{Peak}) were made relative to time-zero for fuel-rich mixtures at an equivalence ratio of $\phi = 2.0$. Temperature was varied from 1412 to 1587 K for 1.5-atm pressure and from 1325 to 1572 K at 10.3 atm. Measurements and model predictions of peak OH* formation times at 1.5 and 10.3 atm are shown in Fig. 5. Excellent agreement is achieved between experimental data and model prediction for peak times at higher pressure, however, the model shows disagreement both quantitatively (over-predicting peak times) and qualitatively (larger activation energy) at lower pressure.

Ignition delay times were unable to be defined from model predictions of OH*(t) profiles due to some difficulty arising from unclear ignition peaks. The discrepancy is more an artifact of exaggerated yield and wider profiles (longer duration) of incipient OH* kinetics, within the first 60–80 μs, and resulting difficulty in assigning a specific ignition delay time at these conditions than problems with the model. The validity of this statement is shown through the comparison of peak OH* formation times which, relative to ignition times, provides a more representative metric for kinetic model predictions at the fuel-rich condition.

Measured activation energy for peak OH* yield increased by 12.1 kcal/mol as reaction pressure increased, with 31.4 and 43.5 kcal/mol calculated for 1.5 and 10.3 atm trends, respectively. Model-predicted activation energy for times to peak OH* yield of 51.7 and 47.3 kcal/mol were calculated for 1.5 and 10.3 atm, respectively. The over-prediction of times at lower pressure resulted in a 65% difference in calculated activation energy. The strong agreement of peak OH* times at 10.3 atm is reflected in a slight difference of 3.8 kcal/mol, a percentage difference of only 8.8%.

4.4. Ignition delay time correlation

An empirical ignition delay time correlation of modified Arrhenius form was constructed using the method of least squares. Utilizing measurements obtained under all conditions of pressure and composition (93 data points), statistics of the line passing through the collapsed data set reveal an overall activation energy for the highly dilute mixtures of $E_A = 51.0$ kcal/mol (25671 K for E_A/R) and a pressure dependence of the form P^n, where $n = -0.48$. The temperature-dependent ignition correlation is placed in terms of molar concentrations of n-nonane and oxygen. Units in Eq. (1) consist of mol/cm^3 and K for species concentration and activation energy/temperature, respectively.

$$\tau(\mu s) = 2.71 \times 10^{-8}[\text{n-}C_9H_{20}]^{0.73}[O_2]^{-1.21} \times \exp(25671/T) \quad (1)$$

Measured ignition delay times agree strongly with those calculated using Eq. (1), where a linear fitting parameter of $R^2 = 0.96$ describes the concurrence (Fig. 6).

5. Model prediction of laminar flame speed

Laminar flame speeds of premixed n-alkane/air flames were measured by Ji et al. [7] in a heated counterflow burner system where Laser Doppler Velocimetry served as the diagnostic. Flame speed results for C_5–C_{12} n-alkanes were calculated utilizing an extrapolation technique to account for non-linear effects due to thermal expansion on reference flame velocity (defined in [7] at 0.5 cm from burner nozzle exit) as the strain rate approaches zero. Laminar flame speeds were calculated using

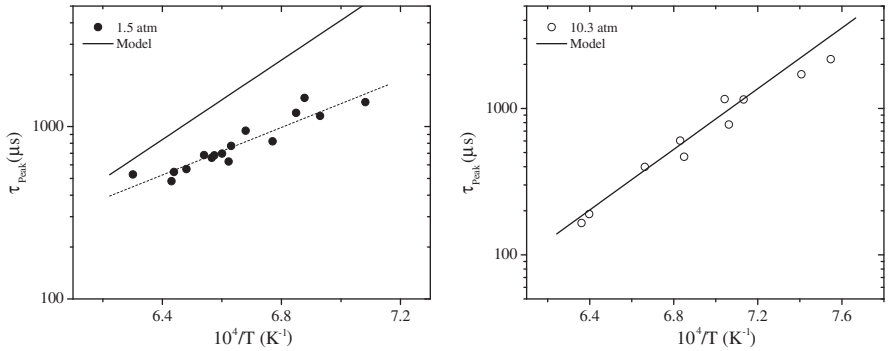

Fig. 5. Measurements and model predictions of times to peak OH* formation at 1.5 atm (Left) and 10.3 atm (Right). $0.001n\text{-}C_9H_{20} + 0.007O_2 + 0.992Ar$ ($\phi = 2.0$).

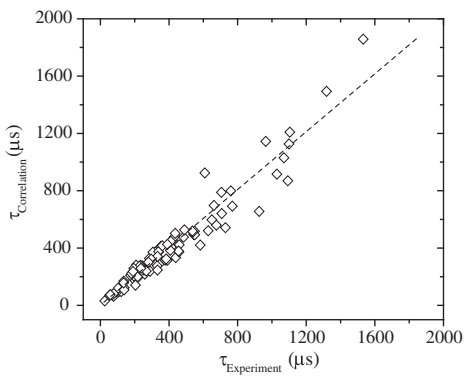

Fig. 6. Comparison of measured ignition delay times to calculated times using Eq. (1).

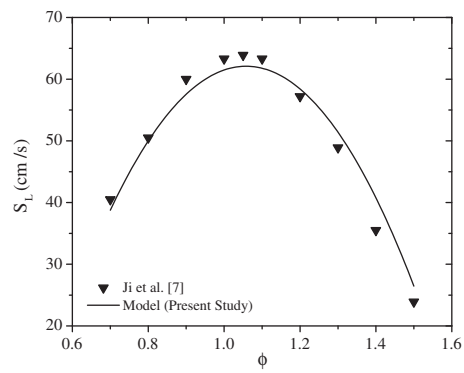

Fig. 7. Model prediction of laminar flame speed measurements of Ji et al. [7]. $P = 1$ atm, $T = 403$ K (unburned gas temperature).

the model described in the present work, conducted using the Premix module in CHEMKIN. Model results are in excellent agreement with the reported *n*-nonane measurements (Fig. 7). Model discrepancy is below 8%, except at the equivalence ratio nearest the rich limit where error approaches 20% ($\phi = 1.5$).

6. Summary

Fundamental measurements and modeling of *n*-nonane oxidation were performed. The proposed mechanism was validated over a wide range of experimental conditions using measurements of transient OH* depletion behind reflected shock waves, ignition delay times, and temperature-dependent product species. The model accurately predicts ignition delay times for lean and stoichiometric conditions; however, at the fuel-rich condition measurements present discrepancies, primarily at high temperatures, due to unclear ignition peaks. To provide a representative metric for kinetic model performance at the rich condition, peak OH* yield times were compared where the model shows excellent agreement at higher pressure. From measurements of oxidation product species, CO_2, CO, H_2O, O_2, CH_4, CH_2O, C_2H_4, and C_3H_6 are the most abundant, and model predictions of these species are in strong agreement with experimental data. Oxygenated hydrocarbons were present for all conditions at higher pressure. At 1 atm, oxygenates were present for lean and stoichiometric conditions, mainly at high temperature, however, aside from CH_2O showed negligible presence at the fuel-rich equivalence ratio. Ethylene showed to be a dominant product of oxidation for all conditions.

Acknowledgment

This work was funded in part by the National Science Foundation, Grant number CBET-0832561.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.055.

References

[1] J.-C. Guibet, *Fuels and Engines*, Editions Technip, Paris, 1999.

[2] P. Dagaut, *Phys. Chem. Chem. Phys.* 4 (2002) 2079–2094.

[3] D. Kunzru, Y.T. Shah, E.B. Stuart, *Ind. Eng. Chem. Process. Des. Dev.* 11 (1972) 605–612.

[4] P. Zhou, O.L. Hollis, B.L. Crynes, *Ind. Eng. Chem. Res.* 26 (1987) 852–855.

[5] J.B. Koffend, N. Cohen, *Int. J. Chem. Kinet.* 28 (1996) 79–87.

[6] C.K. Westbrook, W.J. Pitz, O. Herbinet, H.J. Curran, E.J. Silke, *Combust. Flame* 156 (2009) 181–199.

[7] C. Ji, E. Dames, Y.L. Wang, H. Wang, F.N. Egolfopoulos, *Combust. Flame* 157 (2010) 277–287.

[8] D.F. Davidson, S.C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R.K. Hanson, *J. Prop. Power* 26 (2010) 280–287.

[9] B. Rotavera, E.L. Petersen, in: K. Hannemann, F. Seiler (Eds.), *26th International Symposium on Shock Waves*, Springer-Verlag, Berlin, 2007, pp. 141–146.

[10] J. de Vries, C. Aul, A. Barrett, D. Lambe, E.L. Petersen, in: K. Hannemann, F. Seiler (Eds.), *26th International Symposium on Shock Waves*, Springer-Verlag, Berlin, 2007, pp. 171–176.

[11] P. Dagaut, M. Cathonnet, *Prog. Energy Combust. Sci.* 32 (2006) 48–92.

[12] E.L. Petersen, M.J.A. Rickard, M.W. Crofton, E.D. Abbey, M.J. Traum, D.M. Kalitan, *Meas. Sci. Technol.* 16 (2005) 1716–1729.

[13] H.P. Ramirez, L.K. Hadj-Ali, P. Diévart, G. Moréac, P. Dagaut, *Energy Fuels* 24 (2010) 1668–1676.

[14] F. Buda, R. Bounaceur, V. Warth, P.A. Glaude, R. Fournet, F. Battin-Leclerc, *Combust. Flame* 142 (2005) 170–186.

[15] J.M. Hall, E.L. Petersen, *Int. J. Chem. Kinet.* 38 (2006) 714–724.

[16] A. Burcat, B. Ruscic, (2006). Available from: <http://garfield.chem.elte.hu/Burcat/THERM.DAT>.

[17] C. Muller, V. Michel, G. Schacchi, G.M. Côme, *J. Chim. Phys. Phys. Chim. Biol.* 92 (1995) 1154–1178.

[18] S.W. Benson, *Thermochemical Kinetics: Methods for the Estimation of Thermochemical Data and Rate Parameters*, Wiley, New York, 1976.